

Internship Report

Internship Organization: Shenzhen Shentai Chuangjian Technology Co., Ltd.

Internship Position: Junior Java Development Engineer

Internship Duration: 12/03/2024 – 01/27/2025

Intern: WenFeng Liao

1. Introduction to the Organization

Shenzhen Shentai Chuangjian Technology Co., Ltd., established in 2019, is a high-tech enterprise specializing in cross-border e-commerce logistics services and information technology development. The company is committed to providing professional technical support for enterprises' digital transformation, including customs clearance services and multi-business mode IT services, meeting enterprises' needs in the digital age.

Core Business Areas:

Platform Development: Building efficient and stable cross-border e-commerce IT platforms to support enterprise digital transformation.

Policy Consulting: Assisting enterprises in interpreting cross-border trade policies and regulations to optimize business compliance.

Technical Support: Providing customized technical solutions to enhance system performance and efficiency.

Market Promotion: Helping clients expand into domestic and international markets to boost brand influence.

Outsourcing Services: Offering comprehensive support ranging from technical development to operational assistance.

With 2 invention patents and 21 software copyrights, Shentai Chuangjian maintains technological leadership in cross-border service platforms and intelligent warehouse management systems. Relying on a robust R&D team and innovative products, the company has gradually built a brand presence in the Greater Bay Area, actively expanding into Southeast Asia, Europe, and other international markets, becoming a key force in the cross-border e-commerce industry. Looking ahead, the company will continue to focus on technological innovation, optimize services, deepen domestic and international market deployment, and strive to become a leading company in the cross-border e-commerce sector.



2. Objectives of the Internship

- Apply theoretical knowledge to practical development to enhance hands-on capabilities.
- Familiarize with the system architecture and business processes of cross-border e-commerce enterprises.
- Master teamwork-based development workflows and technical processes.
- Gain project experience to build a solid foundation for future career development.
- Enhance problem-solving skills and cultivate innovative thinking.

3. Internship Activities

- Participated in the functional module development of the cross-border e-commerce supervision system.
- Collaborated with the team in analyzing business requirements.
- Developed code for core functionalities and performed unit testing and optimization.
- Conducted interface debugging with the team to ensure system stability and reliability.
- Authored development and interface documentation for subsequent maintenance.

4. Project Involvement

1. EHU Station Project

Completed Tasks:

- Developed terminal vehicle management functionalities, enabling real-time operation management of vehicles. Displayed vehicle management features on the web interface through frontend tools and implemented real-time updates on vehicle statuses via backend APIs.

2. QSC Station Project

Completed Tasks:

- Developed an API interface for uploading documents for e-commerce enterprises, analyzing existing logic, adjusting parameter formats, and achieving seamless support for the new system.
- Designed an API for uploading vehicle trajectory images, solving field mapping issues and optimizing upload efficiency.
- Developed an API for querying station information, optimizing SQL query logic, and improving data query performance.
- Modified and developed an API for updating reasons for deleting on-site vehicle information, addressing the absence of certain fields in the new system, ensuring correct API invocation, and returning results.



5. Technologies and Tools

Programming Languages:

- Java
- Vue

Framework Selection:

System Environment:

- Java EE 8
- Servlet 3.0
- Apache Maven 3

Primary Frameworks:

- Spring Boot 2.5.6
- Spring Framework 5.3.33
- Spring Security 5.7.12

Persistence Layer:

- Apache MyBatis 3.4.3.4
- Hibernate Validation 6.5.7
- Alibaba Druid 1.2.23

View Layer:

- Vue 2.6.0
- Axios 0.24.0
- Element 2.15.14

Tools:

- Subversion (Code Management)
- MyNative (Data Development Environment Configuration)
- ApiFox (Interface Debugging and Testing)

Database:

- MySQL

6. Internship Achievements

- Gained familiarity with the full-service process of cross-border e-commerce logistics, including customs clearance, logistics, trunk line, and last-mile delivery services.

- Participated in database design and optimization, learning about database indexing and query optimization.



- Mastered techniques for automated test case generation using Postman and ApiFox, reducing manual testing workload.
- Improved documentation skills and became familiar with technical documentation standards and processes.
- Gained proficiency in using Redis for cache design and optimization, including hotspot data caching and distributed lock implementation, enhancing system throughput.
- Mastered the use of API documentation auto-generation tools like Swagger, improving collaboration between development and testing teams.
- Familiarized with the frontend-backend separation development model.

7. Internship Reflections and Gains

1. Key Takeaways:

Through participation in multiple projects, I not only improved my technical capabilities but also gained a comprehensive understanding of the cross-border e-commerce industry. I mastered system architecture and accumulated experience in interface development and system optimization. I also deeply realized the importance of applying theoretical knowledge in practical development and maintaining code standards in complex projects.

2. Challenges and Solutions:

- Challenge: Lack of familiarity with business logic

Solution: Actively researched business materials and communicated with the business team to deepen understanding.

- Challenge: Compatibility issues during interface migration

Solution: Used debugging tools to test interfaces and resolved field mapping and logical differences.

- Challenge: Issues in frontend-backend debugging

Solution: Collaborated with team members to resolve inconsistencies in data transmission.

3. Importance of Team Collaboration:

In various projects, I deeply understood the importance of teamwork. Whether in requirement analysis or interface debugging, efficient communication and clear division of labor among the team ensured the successful completion of complex tasks. I also learned many optimization techniques during team code reviews.

4. Impact on Career Planning:

This internship clarified my future career direction. I aim to continue developing in the cross-border e-commerce and system development fields, focusing on system architecture design and performance optimization. Additionally, I realized the importance of continuous learning and plan to further enhance my technical expertise.



8. Summary and Outlook

This internship not only advanced my technical skills but also cultivated my teamwork and problem-solving abilities. This experience laid a solid foundation for my future career development and gave me a deeper understanding of the integration of technology and business. In the coming studies and work, I will maintain my enthusiasm for learning, continuously improve myself, and create greater value for enterprises.



